

# Will Wu

## Senior Backend Engineer

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### SUMMARY

Senior backend engineer with 4+ years shipping and operating production systems in Go, Python, and Java. Moved a 20-microservice platform onto Alibaba Cloud RDS inside an 11-minute window, then delivered zero-downtime Kubernetes rollout across 21 services, proven live at 7,910 requests with 0 downtime. Cut MongoDB p95 by 80% on a platform serving ~3M weekly API calls. I take systems end to end: architecture, migration, and the incident afterwards.

### WORK EXPERIENCE

#### APMIC — Sr. Backend Engineer

Nov 2025 – Present

Taipei, Taiwan · Python (FastAPI), React, Next.js, PostgreSQL, Hasura, RabbitMQ, Google ADK

Enterprise AI platform for on-premises LLM deployment and model fine-tuning; deliver end-to-end LLM / agent integrations and the full-stack enterprise systems around them for clients.

- **Delivered SAML 2.0 SSO and a privilege-isolating RBAC redesign** for an enterprise customer-service platform (iframe-embedded for a global insurance client) — root-caused a multi-layer SSO failure down to a vendor `xs:type` XSD-validation conflict and shipped a production hot-patch; introduced a type discriminator to cleanly separate JIT-provisioned end-customers from operators, closing a privilege-escalation gap.
- **Built a real-time webchat agent-state system** — a 5-state machine (ready / not-ready / serving / busy / offline) enforced by PostgreSQL triggers, supervisor 1-to-many assignment, and cross-system mutual exclusion against the client's Cisco Finesse telephony platform (fail-open REST gate), covered by unit + Playwright e2e suites.
- **Self-built 3-layer distributed tracing** — contextvar `trace_id` propagated across FastAPI middleware and RabbitMQ message headers, with dual-sink structured logging and PII-safe business-event instrumentation across 6 modules (chosen over OpenTelemetry where no downstream collector existed).
- **Built automated web-scraping pipelines and a RAG knowledge-base integrity toolchain** — manifest-diff verification with exponential-backoff retry that surfaced 64 missing upstream documents and cut actionable gaps from 64 to 1, keeping the LLM retrieval corpus fresh and complete.

#### dailyfresh — Sr. Backend Engineer

Jan 2025 – Present

Part-time · Taipei, Taiwan · Go (Gin), Java (Spring Boot), PostgreSQL, Redis, RabbitMQ, k3s, Next.js, Flutter

Multi-vendor fresh-grocery marketplace (Go / Java microservices + Next.js web + Flutter buyer/seller apps); senior engineer owning backend platform and cross-stack delivery.

- **Re-architected web authentication to a gateway-owned session model** — replaced a BFF / Better-Auth setup with opaque `httpOnly` cookies + CSRF protection and a role-as-claim design (per-request active-role hint validated against the account's role set), unifying buyer/seller dual-role identity; shipped with Playwright e2e regression guards and a single-use Redis-ticket scheme for cookie-authenticated WebSockets.
- **Led a zero-downtime OAuth account-model refactor** from 1:1 columns to a 1:N `auth_providers` table across Google / Apple providers — dual-write → read-cutover → drop-legacy migration with a conflict-resolution flow (`tempBindToken` + link-confirm + last-provider guard) and an in-process Apple-revoke worker (33 commits, no downtime).
- **Owned a production database migration of a 20-microservice platform to Alibaba Cloud RDS** (11 Go + 9 Java services, 100 migrations, 19 pods) within an ~11-minute maintenance window — resolved migration-history, TLS, and Cloudflare-525 edge issues during cutover.
- **Delivered a merchant-governance / food-safety-reporting epic end-to-end in a day** (~16,000 lines across Go / proto / SQL / Next.js) and drove the buyer/seller iOS apps through their first App Store submission (full G0-G7 launch pipeline + root-caused an App Store Connect IRIS bundle-id substring-match bug).

#### Vocus — Sr. Backend Engineer

Mar 2025 – Oct 2025

Taipei, Taiwan · Go (Gin), MongoDB, Redis, Qdrant, Cloudflare, AWS, GCP

- **Contributed to DDD refactoring of Taiwan's top-tier content-creator platform** (CDN-fronted, 94%+ Cloudflare cache-hit; backend handled ~3M weekly API calls for memberships, subscriptions, comments, recommendations, payments) — migrated legacy handlers into a 4-layer architecture across 73+ API modules; designed composition patterns (single-responsibility services + functional decorator chains with concurrent fan-out) to cut god-function complexity.
- **Owned end-to-end migration of the NewwebPay payment integration** from server-side to iframe-redirect mode across all 4 payment types (subscription / one-time / recurring / donation) — moved card-data collection to the gateway's iframe to reduce PCI-DSS scope; redesigned token lifecycle across legacy + new paths.
- **Reduced MongoDB p95 latency on hotspot endpoints by 80%** (platform-wide API p95 80 ms, p50 4 ms) — restructured queries from `$lookup` cross-collection joins to single-collection reads + application-layer joins (goroutine fan-out), leveraging the document model instead of mimicking SQL.
- **Implemented batched async pipelines** — AWS SQS delayed-notification queue (batch 100–500, group-by aggregation, retry with backoff), hourly stats aggregation via worker pool (`InsertMany` replacing per-event writes), and periodic Qdrant-driven recommendation computation for article internal linking (SEO).
- **Migrated the recommendation system from BigQuery to self-hosted Qdrant vector DB, lowering infra cost by 20%** — eliminated per-query usage-based pricing for the highest-traffic query path.

New Taipei, Taiwan · Python (FastAPI), Next.js, Tauri, PostgreSQL, Redis, GCP, Azure

- Joined as the company’s first and only Sr. backend engineer in an early-stage startup; architected and built 9 backend microservices from scratch (unified Python 3.12 / FastAPI / SQLAlchemy 2) covering cloud POS, short-term EV rental, AI meeting summarization, and multiple vertical ERPs.
- Owned all backend architectural decisions, technical selection, and code review; mentored junior engineers as the team grew from 4 to 7–8 people.
- Built development infrastructure from zero — self-hosted GitLab, CI/CD (GitLab CI + Ansible) reducing deployment time by 50%, IaC (Terraform) reducing infrastructure overhead by 30%, SonarQube quality gates.
- Shipped a commercial AI meeting-summarization product solo in 3 months — Whisperx (ASR with speaker diarization) + Ollama (local LLM) + a Tauri desktop app (Next.js 16 / React 19); currently serving paying customers.
- Shipped a cloud POS (F&B + retail) and a short-term EV-rental system (IoT + payment + SMS) — each ~2 months greenfield solo development.

AI-Rider — Backend Engineer

Mar 2022 – Nov 2023

New Taipei, Taiwan · Node.js (Koa), Python (Flask), MariaDB, Redis, RabbitMQ, Modbus TCP

- **Primary backend engineer on a POS system powering ~1,000 on-premise parking sites** (DuDuParking as flagship client, 50%+ of company revenue) — handled multi-tenant billing (time-tiered, monthly pass, EV-charging pre-auth, municipal contract rates).
- **Architected a hardware abstraction layer (ioserver) supporting 10+ device types** via Modbus TCP — ANPR cameras, gate arms, ETag, LED boards — with OOP design and a unified REST API hiding protocol complexity from upstream services.
- **Rebuilt the LED broadcasting system controlling thousands of LED panels across ~1,000 sites** — consolidated a 2-service legacy architecture (Go datasync + Python device backend) into a unified Turbo monorepo (Next.js + Electron + Python Flask), retiring the legacy Go service entirely.
- Optimized SQL queries reducing execution time by 50% for a customer with ~4–5M parking records (EXPLAIN analysis + compound indexes).
- Built a YouTube-based Asian facial dataset pipeline processing 1,000+ videos (multi-stream threading pool, incremental validation) to address facial-recognition model bias.

SIDE PROJECTS

bfx-funding-bot — Algorithmic Crypto Funding Bot (Solo) 2026 · Python (asyncio), PostgreSQL, Koyeb · [case study](#)

- **Built and run a live, real-money Bitfinex margin-funding bot end-to-end** — event-sourced ledger with a periodic-reconcile correctness backbone (REST snapshot as source of truth, WebSocket as best-effort latency optimization, à la FIX drop-copy), single-writer exposure reconciliation, and balance-aware deployment gating; diagnosed and fixed 6+ production incidents on live capital, each shipped with a canary verification step.
- **Designed a quantitative strategy-validation framework** — walk-forward optimization, out-of-sample profitability with bootstrap confidence intervals and deflated Sharpe, and a bot-vs-idle attribution model.

divego — Full-Stack Migration (Solo) 2026 · TypeScript (Next.js, Hono, Drizzle) · [case study](#)

- **Migrated a FastAPI backend to a unified Next.js full-stack app** (Hono + Drizzle on Next.js 16) using the strangler-fig pattern — domain-by-domain vertical slices with a cross-stack unified error protocol, retiring the Python backend entirely (~567 tests).

SKILLS & TECHNICAL COMPETENCIES

|                    |                                                                            |
|--------------------|----------------------------------------------------------------------------|
| Languages          | Python, Go, TypeScript, Dart, Java                                         |
| Backend            | FastAPI, Gin, Koa, Flask, Spring Boot, Hono                                |
| Frontend / Desktop | Next.js, React, Tauri, Flutter                                             |
| Databases          | PostgreSQL, MariaDB, MongoDB, Redis, Qdrant (vector DB)                    |
| Data / Migration   | SQLAlchemy, Drizzle, Atlas, Alembic, Flyway                                |
| Messaging          | RabbitMQ, MQTT                                                             |
| AI / ML            | Whisperx (ASR), Ollama (local LLM), Google ADK, PyTorch / CUDA             |
| Cloud              | GCP, AWS, Azure, Cloudflare, Alibaba Cloud                                 |
| DevOps / Infra     | Docker, k3s, GitLab CI, Terraform, Ansible, Self-hosted GitLab             |
| Protocols          | REST, gRPC, GraphQL (Hasura), SAML / OAuth, Modbus TCP, JSON-RPC, Webhooks |

EDUCATION

MS, Data Analytics Engineering — George Mason University

BS, Chemical Engineering — Tatung University

2019 – 2021

2013 – 2016